

Coordinate-Free Expansion of Multivector Exponentials in $Cl(p, q)$ for $p + q = 3$

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- 1 Introduction
- 2 Exponential formulas with examples
- 3 Trigonometric and hyperbolic functions
- 4 Conclusions and perspective

Well-known exponential formulas in mathematics

Complex number algebra

is isomorphic to $Cl_{0,1}$ $i \equiv \mathbf{e}_1; i^2 = \mathbf{e}_1^2 = -1$; de Moivre's (1667-1754) rule

$$e^{i\phi} = (\cos \phi + i \sin \phi) \rightarrow \underbrace{e^{\mathbf{e}_1 \phi} = (\cos \phi + \mathbf{e}_1 \sin \phi)}_{Cl_{0,1}}$$

Quaternionic algebra, Hamilton (1805-1865)

is isomorphic to $Cl_{0,2}$.

Quaternion: $Q = q_0 + q_1 i + q_2 j + q_3 k = q_0 + \mathbf{q} = \text{scalar} + \text{vector}$.

$$e^Q = e^{q_0} \left(\cos |\mathbf{q}| + \frac{\mathbf{q}}{|\mathbf{q}|} \sin |\mathbf{q}| \right) \quad |\mathbf{q}| = \sqrt{q_1^2 + q_2^2 + q_3^2}$$

Well-known exponential formulas in mathematics

Geometric algebra (GA); A is a multivector (MV)

Mainly trigonometric/hyperbolic Moivre-type formulas are known:

$$e^A = \cos |A| + \frac{A}{|A|} \sin |A| \quad \text{if } A^2 < 0$$

$$e^A = \cosh |A| + \frac{A}{|A|} \sinh |A| \quad \text{if } A^2 > 0,$$

where $|A|$ is the MV magnitude. For $n = p + q = 3$, the characteristic polynomials of matrix representations of $Cl_{p,q}$ are of degree 4. Since the algebraic equations of degree 4 are solvable in radicals, the question arises why the formulas for MV exponentials are still absent?

Motivation and main results

1. To find a general formula for exponential of MV A for $n = 3$.
2. Demonstrate the implementation.

General properties of GA exponential

Some algebraic properties of exponential

$$\exp(A + B) = \exp(A) \exp(B) \quad \text{if and only if } AB = BA,$$

$$\forall \exp(A) V^{-1} = \exp(VAV^{-1}).$$

In the last expression the transformation V , for example the rotor, has been lifted to exponent.

Defining property

$$\left. \frac{\partial \exp(At)}{\partial t} \right|_{t=1} = A \exp(A) = \exp(A)A, \quad \exp(0) = 1.$$

Here A is assumed to be **independent** of t . Since we have a single MV that always commutes with itself the multiplications from left and right by A coincide. This property **can be used to prove** all explicit exponential formulas presented below.

General properties of GA exponential

Action of main involutions

A general MV that belongs to real GAs $Cl_{p,q}$ for $n = p + q = 3$

$$A = a_0 + \underbrace{a_1\mathbf{e}_1 + a_2\mathbf{e}_2 + a_3\mathbf{e}_3}_{\text{vector } \mathbf{a}} + \underbrace{a_{12}\mathbf{e}_{12} + a_{23}\mathbf{e}_{23} + a_{13}\mathbf{e}_{13}}_{\text{bivector } \mathcal{A}} + a_{123}I$$

$$\widetilde{A} = a_0 + \mathbf{a} - \mathcal{A} - a_{123}I, \quad \text{reversion}$$

$$\widehat{A} = a_0 - \mathbf{a} + \mathcal{A} - a_{123}I, \quad \text{grade inversion}$$

$$\widehat{\widetilde{A}} = a_0 - \mathbf{a} - \mathcal{A} + a_{123}I \quad \text{Clifford conjugate}$$

Involutions of exponentials

$$\widetilde{e^A} = e^{\widetilde{A}}, \quad \widehat{e^A} = e^{\widehat{A}}, \quad \widehat{\widetilde{e^A}} = e^{\widehat{\widetilde{A}}}.$$

General properties of GA exponential

Series expansions

P. Lounesto, *Clifford Algebra and Spinors*, 1997.

$$e^A = 1 + \frac{A}{1!} + \frac{A^2}{2!} + \frac{A^3}{3!} + \frac{A^4}{4!} + \dots, \quad \text{Standard form}$$

$$e^A = 1 + \frac{A}{1} \left(1 + \frac{A}{2} \left(1 + \frac{A}{3} \left(1 + \frac{A}{4} (1 + \dots) \right) \right) \right), \quad \text{Horner's rule}$$

Convergence radius is **infinite**. Convergence is **not monotonic**: coefficients may acquire **very large values** before starting to converge.

Example of Horner's rule implementation in *Mathematica*:

```
expMVHorner[A_, n_] := Module[{B = 1, s = n + 1},
  While[(s = s - 1) > 0, B = 1. + GP[B, A/s]]; B,
```

Here n is the number of iterations and GP is the geometric product. If MV coefficients are large ($a_j \geq 3$) to accelerate the convergence the formula $(\exp(A/m))^m$, where m is the integer, may be applied at first and then to raise the result to the m -th power.

General properties of GA exponential

Numerical experiment

$$A = -8 - 6e_2 - 9e_3 + 5e_{12} - 5e_{13} + 6e_{23} - 4e_{123}; \quad a_{12} = 5$$

Horner expansion

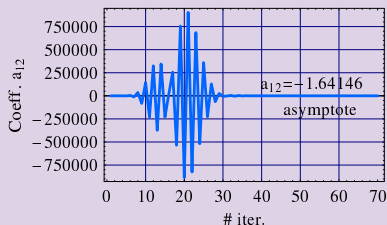


Figure: $Cl_{3,0}$. Alternating bell-like convergence. Dependence of coeff. a_{12} on a number of terms (iterations) in a finite GA exponential series. 20 terms give $a_{12} = 0.902645 \times 10^6$. 70 terms give $a_{12} = -1.64146$ which is close to exact. Similar behavior is observed for remaining coefficients.

Horner with division by largest coeff.

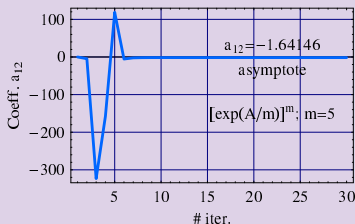


Figure: $Cl_{3,0}$. Acceleration of convergence by $\exp A = (\exp(A/m))^m$. Dependence of a_{12} on a number of terms (iterations) in a finite exponential series. Now 30 terms give correct value $a_{12} = -1.64146$. The max amplitude has decreased 3×10^4 times. Similar behavior is observed for remaining coefficients.

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Exact exponentials for $Cl_{p,q}$ when $p + q = 3$

Coordinate forms for exponentials of $Cl_{3,0}$, $Cl_{1,2}$, $Cl_{2,1}$, and $Cl_{0,3}$ algebras are presented in

A. Dargys and A. Acus, Exponential of general multivector (MV) in 3D Clifford algebras, Nonlinear Analysis: Modelling and Control **26** (2021), (arXiv:2104.01905).

Here we have transformed the exponentials into **coordinate-free** generic formulas

Notation

$$A = a_0 + \underbrace{a_1\mathbf{e}_1 + a_2\mathbf{e}_2 + a_3\mathbf{e}_3}_{\text{vector } \mathbf{a}} + \underbrace{a_{12}\mathbf{e}_{12} + a_{23}\mathbf{e}_{23} + a_{13}\mathbf{e}_{13}}_{\text{bivector } \mathcal{A}} + a_{123}I$$

$Cl_{0,3}$ algebra $Cl_{0,3}$ generic formula

$$\exp(A) = \frac{1}{2} e^{a_0} \left(e^{a_{123}} (1 + I) \left(\cos a_+ + \frac{\sin a_+}{a_+} (\mathbf{a} + \mathcal{A}) \right) + e^{-a_{123}} (1 - I) \left(\cos a_- + \frac{\sin a_-}{a_-} (\mathbf{a} + \mathcal{A}) \right) \right), \quad (1)$$

where a_- and a_+ are the scalars,

$$\begin{aligned} a_- &= \sqrt{-(\mathbf{a} \cdot \mathbf{a} + \mathcal{A} \cdot \mathcal{A}) + 2\mathbf{a} \wedge \mathcal{A}} \\ a_+ &= \sqrt{-(\mathbf{a} \cdot \mathbf{a} + \mathcal{A} \cdot \mathcal{A}) - 2\mathbf{a} \wedge \mathcal{A}}. \end{aligned} \quad (2)$$

- Vector and bivector components are mixed up: $a_- = \sqrt{(a_3 + a_{12})^2 + (a_2 - a_{13})^2 + (a_1 + a_{23})^2}$, $a_+ = \sqrt{(a_3 - a_{12})^2 + (a_2 + a_{13})^2 + (a_1 - a_{23})^2}$.
- When the denominator in the formula (1), either a_+ or a_- goes to zero we will have a **special case**. The generic formula (1) then should be modified by replacing the corresponding ratios by their limits $\lim_{a_{\pm} \rightarrow 0} \frac{\sin a_{\pm}}{a_{\pm}} = 1$.

$C_{0,3}$ algebra

Example 1

The exponential of $A = -8 - 6\mathbf{e}_2 - 9\mathbf{e}_3 + 5\mathbf{e}_{12} - 5\mathbf{e}_{13} + 6\mathbf{e}_{23} - 4\mathbf{e}_{123}$. Using (1) we find $a_- = \sqrt{53}$ and $a_+ = \sqrt{353}$. The exact answer is

$$\begin{aligned} \exp(A) = & \frac{e^{-8}}{2} \left(e^4 (1 - \mathbf{e}_{123}) \right. \\ & \times \left(\cos \sqrt{53} + \frac{\sin \sqrt{53}}{\sqrt{53}} (-6\mathbf{e}_2 - 9\mathbf{e}_3 + 5\mathbf{e}_{12} - 5\mathbf{e}_{13} + 6\mathbf{e}_{23}) \right) \\ & + e^{-4} (1 + \mathbf{e}_{123}) \\ & \times \left. \left(\cos \sqrt{353} + \frac{\sin \sqrt{353}}{\sqrt{353}} (-6\mathbf{e}_2 - 9\mathbf{e}_3 + 5\mathbf{e}_{12} - 5\mathbf{e}_{13} + 6\mathbf{e}_{23}) \right) \right). \end{aligned}$$

The calculation of the exponential series using Mathematica's floating point arithmetic gives **six significant figures** of the exact solution after summation of **70 series terms**. The convergence is not monotonic. To get the **first significant figure** for all basis MV elements no less than **64 standard series terms** (iterations) are needed in this particular case.

$Cl_{3,0}$ and $Cl_{1,2}$ algebras

$Cl_{3,0}$ and $Cl_{1,2}$ generic formula

$$\begin{aligned} \exp(A) = & e^{a_0} (\cos a_{123} + I \sin a_{123}) \left(\cos a_- \cosh a_+ + I \sin a_- \sinh a_+ \right. \\ & + \frac{1}{a_+^2 + a_-^2} (\cosh a_+ \sin a_- - I \cos a_- \sinh a_+) \\ & \left. \times (a_- (\mathbf{a} + \mathcal{A}) + a_+ I (\mathbf{a} + \mathcal{A})) \right), \end{aligned} \quad (3)$$

where scalars $a_{\pm}$ are

$$\begin{aligned} a_- &= \frac{-2/\mathbf{a} \wedge \mathcal{A}}{\sqrt{2} \sqrt{\mathbf{a} \cdot \mathbf{a} + \mathcal{A} \cdot \mathcal{A} + \sqrt{(\mathbf{a} \cdot \mathbf{a} + \mathcal{A} \cdot \mathcal{A})^2 - 4(\mathbf{a} \wedge \mathcal{A})^2}}}, \\ a_+ &= \frac{\sqrt{\mathbf{a} \cdot \mathbf{a} + \mathcal{A} \cdot \mathcal{A} + \sqrt{(\mathbf{a} \cdot \mathbf{a} + \mathcal{A} \cdot \mathcal{A})^2 - 4(\mathbf{a} \wedge \mathcal{A})^2}}{\sqrt{2}} \end{aligned} \quad (4)$$

when $\mathbf{a} \wedge \mathcal{A} \neq 0$, and (see next slide)

$Cl_{3,0}$ and $Cl_{1,2}$ algebras $Cl_{3,0}$ and $Cl_{1,2}$ generic formula (continuation)

$$\begin{cases} a_+ = \sqrt{\mathbf{a} \cdot \mathbf{a} + \mathcal{A} \cdot \mathcal{A}}, & a_- = 0, & \mathbf{a} \cdot \mathbf{a} + \mathcal{A} \cdot \mathcal{A} > 0 \\ a_+ = 0, & a_- = \sqrt{-(\mathbf{a} \cdot \mathbf{a} + \mathcal{A} \cdot \mathcal{A})}, & \mathbf{a} \cdot \mathbf{a} + \mathcal{A} \cdot \mathcal{A} < 0, \end{cases}$$

when $\mathbf{a} \wedge \mathcal{A} = 0$.

When $\text{Det}(\mathbf{a} + \mathcal{A}) = (a_+^2 + a_-^2)^2 = 0$ we have **special case**
 $\exp(\mathbf{A}) = e^{a_0}(\cos a_{123} + I \sin a_{123})$ which can be obtained by
 computing limit of (3), when both $a_+ \rightarrow 0$ and $a_- \rightarrow 0$. Moivre-type
 formulas which follow from Eq. (3) are

$$\begin{aligned} \exp \mathbf{A} &= e^{a_0}(\cos a_{123} + \sin a_{123} I) & \mathbf{a} = \mathcal{A} = 0, \\ \exp \mathcal{A} &= \cos|\mathcal{A}| + \frac{\mathcal{A}}{|\mathcal{A}|} \sin|\mathcal{A}| & a_0 = a_{123} = \mathbf{a} = 0, \\ \exp \mathbf{a} &= \cosh|\mathbf{a}| + \frac{\mathbf{a}}{|\mathbf{a}|} \sinh|\mathbf{a}| & a_0 = a_{123} = \mathcal{A} = 0. \end{aligned}$$

$Cl_{3,0}$ and $Cl_{1,2}$ algebras

Example 2: $Cl_{3,0}$ case

$A = -8 - 6\mathbf{e}_2 - 9\mathbf{e}_3 + 5\mathbf{e}_{12} - 5\mathbf{e}_{13} + 6\mathbf{e}_{23} - 4\mathbf{e}_{123}$. Using (3) we find $\mathbf{a} \cdot \mathbf{a} + \mathcal{A} \cdot \mathcal{A} = 31$, $-2/\mathbf{a} \wedge \mathcal{A} = -150$. Then $a_- = -75\sqrt{\frac{2}{31+\sqrt{23461}}}$ and $a_+ = \sqrt{\frac{31+\sqrt{23461}}{2}}$. The exact numerical answer therefore is

$$\begin{aligned} \exp(A) = & \frac{1}{e^8} (\cos(4) - \sin(4)I) \left(\cos\left(75\sqrt{\frac{2}{31+\sqrt{23461}}}\right) \cosh\left(\sqrt{\frac{31+\sqrt{23461}}{2}}\right) \right. \\ & \left. - \sin\left(75\sqrt{\frac{2}{31+\sqrt{23461}}}\right) \sinh\left(\sqrt{\frac{31+\sqrt{23461}}{2}}\right) I \right) \\ & + \frac{1}{\sqrt{23461}} \left(\left(-75\sqrt{\frac{2}{31+\sqrt{23461}}} (-6\mathbf{e}_2 - 9\mathbf{e}_3 + 5\mathbf{e}_{12} - 5\mathbf{e}_{13} + 6\mathbf{e}_{23}) \right. \right. \\ & \left. \left. + \sqrt{\frac{31+\sqrt{23461}}{2}} (-6\mathbf{e}_2 - 9\mathbf{e}_3 + 5\mathbf{e}_{12} - 5\mathbf{e}_{13} + 6\mathbf{e}_{23}) I \right) \right. \\ & \times \left(-\sin\left(75\sqrt{\frac{2}{31+\sqrt{23461}}}\right) \cosh\left(\sqrt{\frac{31+\sqrt{23461}}{2}}\right) \right. \\ & \left. \left. - \cos\left(75\sqrt{\frac{2}{31+\sqrt{23461}}}\right) \sinh\left(\sqrt{\frac{31+\sqrt{23461}}{2}}\right) I \right) \right). \end{aligned}$$

$Cl_{3,0}$ and $Cl_{1,2}$ algebras

Example 3: $Cl_{1,2}$ case

Let's take a simple MV, $A = 3 - \mathbf{e}_1 + 2\mathbf{e}_{12}$, which represents disentangled case $\mathbf{a} \wedge \mathcal{A} = 0$. Then we have $a_+ = \sqrt{5}$ and $a_- = 0$. The answer is expressed in **hyperbolic** functions:

$$\exp(A) = e^3 \left(\cosh \sqrt{5} + (-\mathbf{e}_1 + 2\mathbf{e}_{12}) \frac{\sinh \sqrt{5}}{\sqrt{5}} \right).$$

$Cl_{2,1}$ algebra

$Cl_{2,1}$ generic formula

The formula is very similar to $Cl_{0,3}$ case.

$$\exp(A) = \frac{1}{2} e^{a_0} \left(e^{a_{123}} (1 + I) (\text{co}(a_+^2) + \text{si}(a_+^2)(\mathbf{a} + \mathcal{A})) \right. \\ \left. + e^{-a_{123}} (1 - I) (\text{co}(a_-^2) + \text{si}(a_-^2)(\mathbf{a} + \mathcal{A})) \right) \quad (5)$$

$$a_-^2 = -(\mathbf{a} \cdot \mathbf{a} + \mathcal{A} \cdot \mathcal{A}) + 2I \mathbf{a} \wedge \mathcal{A}, \quad a_-^2 \geq 0,$$

$$a_+^2 = -(\mathbf{a} \cdot \mathbf{a} + \mathcal{A} \cdot \mathcal{A}) - 2I \mathbf{a} \wedge \mathcal{A}, \quad a_+^2 \geq 0.$$

Functions si and co reduces to **trigonometric or hyperbolic** functions:

$$\text{si}(a_+^2) = \begin{cases} \frac{\sin \sqrt{a_+^2}}{\sqrt{a_+^2}}, & a_+^2 > 0 \\ \frac{\sinh \sqrt{-a_+^2}}{\sqrt{-a_+^2}}, & a_+^2 < 0 \end{cases}; \quad \text{co}(a_+^2) = \begin{cases} \cos \sqrt{a_+^2}, & a_+^2 > 0 \\ \cosh \sqrt{-a_+^2}, & a_+^2 < 0 \end{cases}$$

$Cl_{2,1}$ algebra

$Cl_{2,1}$ generic formula (continuation)

and

$$\text{si}(a_-^2) = \begin{cases} \frac{\sin \sqrt{a_-^2}}{\sqrt{a_-^2}}, & a_-^2 > 0 \\ \frac{\sinh \sqrt{-a_-^2}}{\sqrt{-a_-^2}}, & a_-^2 < 0 \end{cases}; \quad \text{co}(a_-^2) = \begin{cases} \cos \sqrt{a_-^2}, & a_-^2 > 0 \\ \cosh \sqrt{-a_-^2}, & a_-^2 < 0 \end{cases}$$

Notation $a_{\pm}^2 < 0$ is a bit **unusual**. We wanted to keep the same notations as in $Cl_{3,0}$ case, where the $a_{\pm}^2$ is always positively definite. Depending on signs, we have four cases to select.

When $a_- = 0$ and/or $a_+ = 0$ we have **special case**, which again can be straightforwardly included taking already mentioned limits, i.e., by putting $\text{co}(0) = 1$ and $\text{si}(0) = 1$.

$Cl_{2,1}$ algebra

Example 4: case $a_-^2 < 0$, $a_+^2 > 0$.

Using the same MV $A = -8 - 6\mathbf{e}_2 - 9\mathbf{e}_3 + 5\mathbf{e}_{12} - 5\mathbf{e}_{13} + 6\mathbf{e}_{23} - 4\mathbf{e}_{123}$ for $Cl_{2,1}$ we have $a_-^2 = -141$, $a_+^2 = 159$. The answer then is

$$\begin{aligned} \exp(A) = \frac{1}{2e^8} & \left(\frac{1}{e^4} (1 + I) \left(\frac{\sin \sqrt{159}}{\sqrt{159}} (-6\mathbf{e}_2 - 9\mathbf{e}_3 + 5\mathbf{e}_{12} - 5\mathbf{e}_{13} + 6\mathbf{e}_{23}) \right. \right. \\ & \left. \left. + \cos \sqrt{159} \right) \right. \\ & \left. + e^4 (1 - I) \left(\frac{\sinh \sqrt{141}}{\sqrt{141}} (-6\mathbf{e}_2 - 9\mathbf{e}_3 + 5\mathbf{e}_{12} - 5\mathbf{e}_{13} + 6\mathbf{e}_{23}) \right. \right. \\ & \left. \left. + \cosh \sqrt{141} \right) \right). \end{aligned}$$

$Cl_{2,1}$ algebra

Example 5: case $a_-^2 < 0$, $a_+^2 < 0$.

Exponentiating $A = -6\mathbf{e}_2 + 5\mathbf{e}_{12} + \mathbf{e}_{123}$ of $Cl_{2,1}$ we have $a_-^2 = -11$, $a_+^2 = -11$. The answer then is

$$\exp(A) = \frac{1}{2} (e(1 + I) + e^{-1}(1 - I)) \left(\frac{\sinh \sqrt{11}}{\sqrt{11}} (-6\mathbf{e}_2 + 5\mathbf{e}_{12}) + \cosh \sqrt{11} \right).$$

Example 6: case $a_-^2 > 0$, $a_+^2 > 0$.

Exponentiating $A = 2 + \mathbf{e}_3 + 6\mathbf{e}_{12} + 3\mathbf{e}_{123}$ of $Cl_{2,1}$ we have $a_-^2 = 49$, $a_+^2 = 25$. The answer then is

$$\begin{aligned} \exp(A) = \frac{e^2}{2} & \left(e^3(1 + I) \left(\frac{\sin 5}{5} (\mathbf{e}_3 + 6\mathbf{e}_{12}) + \cos 5 \right) \right. \\ & \left. + e^{-3}(1 - I) \left(\frac{\sin 7}{7} (\mathbf{e}_3 + 6\mathbf{e}_{12}) + \cos 7 \right) \right). \end{aligned}$$

$Cl_{2,1}$ algebra

Example 7: case $a_-^2 > 0$, $a_+^2 < 0$.

Exponentiating $A = 2 - 10\mathbf{e}_2 - 10\mathbf{e}_3 + 2\mathbf{e}_{13} + \mathbf{e}_{23} + \mathbf{e}_{123}$ of $Cl_{2,1}$ we have $a_-^2 = 35$, $a_+^2 = -45$. The answer then is

$$\exp(A) = \frac{e^2}{2} \left(e(1 + I) \left(\frac{\sinh 3\sqrt{5}}{3\sqrt{5}} (-10\mathbf{e}_2 - 10\mathbf{e}_3 + 2\mathbf{e}_{13} + \mathbf{e}_{23}) + \cosh(3\sqrt{5}) \right) + e^{-1}(1 - I) \left(\frac{\sin \sqrt{35}}{\sqrt{35}} (-10\mathbf{e}_2 - 10\mathbf{e}_3 + 2\mathbf{e}_{13} + \mathbf{e}_{23}) + \cos \sqrt{35} \right) \right).$$

Application example

Some of differential MV equations, for example,

$$\frac{dX(t)}{dt} = AX + XB, \quad X(0) = X_0,$$

have solutions that consist of products of GA exponentials:

$$X(t) = e^{tA} X_0 e^{tB}.$$

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Relation to trigonometric/hyperbolic functions

Trigonometric relations

GA trigonometric functions, in general, only exist for real GAs that has a commutative pseudoscalar with property $I^2 = -1$, i.e., only for Clifford algebras $Cl_{3,0}$ and $Cl_{1,2}$ [Chappell2015]. To define them for the algebras $Cl_{0,3}$ and $Cl_{2,1}$ we have to introduce imaginary unit.

$$\sin A = \begin{cases} \frac{I}{2}(e^{-IA} - e^{IA}) & \text{for } Cl_{3,0}, Cl_{1,2}, \\ \frac{i}{2}(e^{-iA} - e^{iA}) & \text{for } Cl_{0,3}, Cl_{2,1}, \end{cases}$$

$$\cos A = \begin{cases} \frac{1}{2}(e^{-IA} + e^{IA}) & \text{for } Cl_{3,0}, Cl_{1,2}, \\ \frac{1}{2}(e^{-iA} + e^{iA}) & \text{for } Cl_{0,3}, Cl_{2,1}, \end{cases}$$

The hyperbolic GA functions do not require imaginary unit:

$$\sinh A = \frac{1}{2}(e^A - e^{-A}), \quad \cosh A = \frac{1}{2}(e^A + e^{-A}).$$

It follows that **all single argument identities remain valid** for $n = 3$ Clifford algebras.

Trigonometric function computation example

The determinant norm

The knowledge of exact formulas allows to investigate the rate of convergence of GA trigonometric and hyperbolic series. Take $Cl_{3,0}$ MV

$$A' = \frac{1}{N} (4 + \mathbf{e}_1 + 3\mathbf{e}_2 - 5\mathbf{e}_3 + 10\mathbf{e}_{12} + 9\mathbf{e}_{13} - 9\mathbf{e}_{23} - 4I), \quad I = \mathbf{e}_{123}.$$

To compare with series expansion formula choose the factor N , which ensures convergence of $\tan A$ series. It become convergent if we chose N by (at least) one larger than the determinant norm. The **determinant norm** is defined as **determinant** of A raised to fractional power $1/k$, where $k = 2^{\lceil n/2 \rceil}$, i.e. $|A| = (\text{Det}(A))^{1/k}$. It can be interpreted as a number of MVs A in a MV product needed to define $\text{Det}(A)$. In 3D case $\text{Det}(A) = A\tilde{A}\tilde{A}\tilde{A}$, therefore, $k = 2^{\lceil 3/2 \rceil} = 2^2 = 4$ and the determinant norm is $|A| = \sqrt[4]{\text{Det}(A)}$. For the chosen MV A' we find $\text{Det}(A') = 71129$ and $|A'| = \sqrt[4]{71129} \approx 16.33$. The determinant norm factor $1/|A|$ ensures, that determinants of series terms make a decreasing sequence, $|\text{Det}(A)| > |\text{Det}(AA)| > \dots > |\text{Det}(AA \dots A)|$.

Trigonometric formulas

Exact trigonometric results for $A'' = \frac{1}{17}A'$

Take $Cl_{3,0}$ algebra for which $I^2 = -1$. $\sin A$ and $\cos A$ are expressed as sum of exponents. $\tan A = \sin(A)(\cos A)^{-1}$. Inverse MV $(\cos A)^{-1}$ can be computed as $A^{-1} = \frac{\text{Adj}(A)}{\text{Det}(A)}$, where $\text{Adj}(A)A = A\text{Adj}(A) = \text{Det}(A)$. The numerical MV A'' was inserted to find the following **exact** GA functions presented below with 8 significant figures,

$$\begin{aligned}\sin A'' &= 0.4142215 + 0.1561775\mathbf{e}_1 + 0.2887099\mathbf{e}_2 - 0.4312306\mathbf{e}_3 \\ &\quad + 0.6127064\mathbf{e}_{12} + 0.5664210\mathbf{e}_{13} - 0.5864014\mathbf{e}_{23} - 0.2430952I, \\ \cos A'' &= 1.3837580 + 0.1075001\mathbf{e}_1 + 0.0726490\mathbf{e}_2 - 0.0516785\mathbf{e}_3 \\ &\quad - 0.2436586\mathbf{e}_{12} - 0.1984718\mathbf{e}_{13} + 0.1707105\mathbf{e}_{23} + 0.4152926I, \\ \tan A'' &= 0.0520468 - 0.0321336\mathbf{e}_1 + 0.0568865\mathbf{e}_2 - 0.1373908\mathbf{e}_3 \\ &\quad + 0.4876809\mathbf{e}_{12} + 0.4261388\mathbf{e}_{13} - 0.4091069\mathbf{e}_{23} - 0.1473168I.\end{aligned}$$

Trigonometric formulas

Standard series trigonometric results for $A'' = \frac{1}{17}A'$

Using **series expansion** of $\sin A$, $\cos A$ and $\tan A$, we find,

$$\sin_6 A'' = 0.4141938 + 0.1560854\mathbf{e}_1 + 0.2886852\mathbf{e}_2 - 0.4312611\mathbf{e}_3 \\ + 0.6131181\mathbf{e}_{12} + 0.5667705\mathbf{e}_{13} - 0.5867229\mathbf{e}_{23} - 0.2437297I,$$

$$\cos_6 A'' = 1.3838520 + 0.1075543\mathbf{e}_1 + 0.0727025\mathbf{e}_2 - 0.0517373\mathbf{e}_3 \\ - 0.2436926\mathbf{e}_{12} - 0.1984933\mathbf{e}_{13} + 0.1707199\mathbf{e}_{23} + 0.4153298I,$$

$$\tan_6 A'' = 0.0958579 + 0.0035747\mathbf{e}_1 + 0.0832419\mathbf{e}_2 - 0.1588803\mathbf{e}_3 \\ + 0.4184797\mathbf{e}_{12} + 0.3705885\mathbf{e}_{13} - 0.3625310\mathbf{e}_{23} - 0.0454115I.$$

$$\tan_{40} A'' = 0.0522097 - 0.0320415\mathbf{e}_1 + 0.0569781\mathbf{e}_2 - 0.1374922\mathbf{e}_3 \\ + 0.4876273\mathbf{e}_{12} + 0.4261060\mathbf{e}_{13} - 0.4090946\mathbf{e}_{23} - 0.1472580I.$$

The subscripts at trigonometric functions show the number of terms used in series expansion. Comparison with the exact results ([previous slide](#)) shows that three significant figures are correct. Note **difference** between $\tan_6 A$ and $\tan_{40} A$.

- 1 Introduction
- 2 Exponential formulas with examples
- 3 Trigonometric and hyperbolic functions
- 4 Conclusions and perspective**

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- The exponential functions are needed in solving GA differential equations, in signal processing, automatic control and robotics.

Implementation of exponential formulas

We implemented the described formulas of the exact exponents (both in coordinate and coordinate free form) in *Mathematica* package, which may be download from <https://github.com/ArturasAcus/GeometricAlgebra>.

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